

Mathematics for Social Scientists

Part 2: Introduction to Analysis Integrals

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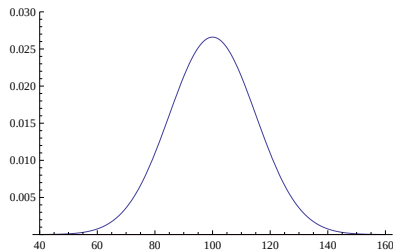
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Motivation (Statistics)

- Probability density functions (p.d.f.) are fundamental in statistics.
- To get the probability of a *range* of outcomes, we need the **area under the p.d.f.**
- Example: $\text{IQ} \sim \mathcal{N}(100, 15^2)$. What is $P(X > 120)$?

$$P(X > 120) = \int_{120}^{\infty} f_X(x) dx \approx 0.0912$$



Motivation: Policy & Demography

- **Political Science (mobilization impact over time).** Net effect on $[t_0, t_1]$ is the accumulated *rate* difference:

$$\text{Net effect} = \int_{t_0}^{t_1} \left(r_{\text{policy}}(t) - r_{\text{cf}}(t) \right) dt = \left[y_{\text{policy}}(t) - y_{\text{cf}}(t) \right]_{t_0}^{t_1}$$

where r is a flow/rate (e.g., *protest participants per hour*, *voters contacted per day*) and $y'(t) = r(t)$.

- **Sociology (demography & exposure).** Totals from time-varying rates:

$$\text{Total migrants} = \int_0^T m(t) dt$$

Person-time and cumulative exposure (dose):

$$\text{Person-time} = \int_0^T N(t) dt, \quad \text{Cumulative exposure} = \int_0^T c(t) dt$$

with $N(t)$ the population under observation and $c(t)$ a concentration (pollution/media).

Motivation: Psychology & Behavioral Time Courses

- **Area-under-the-curve (AUC).** Summarize response trajectories by total response:

$$\text{AUC} = \int_{t_0}^{t_1} r(t) dt \quad (\text{or above baseline: } \int_{t_0}^{t_1} [r(t) - r_0] dt)$$

where $r(t)$ might be cortisol/hormone level, pupil dilation, or BOLD/EEG signal.

- **Learning/engagement curves.** Total engagement over a session:

$$\text{Total engagement} = \int_0^T e(t) dt$$

(e.g., attention/accuracy over time in a task, adherence in interventions, app-usage intensity).

Common thread: integrate *rates/flows* to get *totals/levels*.

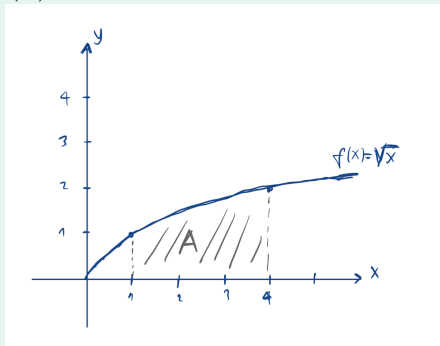
Areas, Sums, Integrals

Intuition: Areas, Sums, Integrals

We want a neat way to calculate the area that lies “under the function.”

Example.

Calculate the area under $f(x) = \sqrt{x}$ from 1 to 4:



Intuition: Integrals and Areas

We can approximate the area:

1. Divide the interval into h equally spaced bars (rectangles),
2. each bar has width $\Delta x = \frac{b-a}{h}$ and height sampled from the curve,
3. add up all bar areas.

Figures next show two simple sampling choices, with $h = 3$ bars.

Intuition: Integrals and Areas

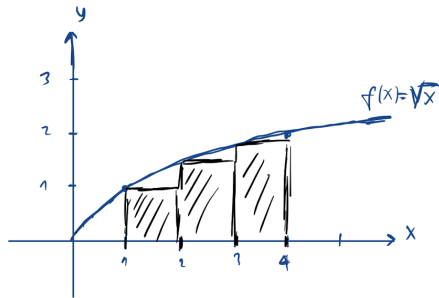


Figure: $\Delta x \sum_{i=0}^{h-1} f(a + i \Delta x)$
(Left Riemann sum)

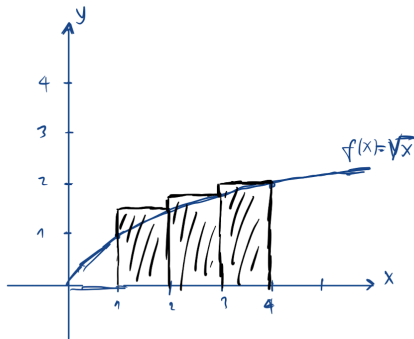


Figure: $\Delta x \sum_{i=1}^h f(a + i \Delta x)$
(Right Riemann sum)

- Δx is the bar width; h is the number of bars.
- Left (right) sum samples the **leftmost (rightmost)** endpoint of each subinterval.

Intuition: Areas, Sums, Integrals

- **Remark:** More bars $\Rightarrow$ thinner bars (Δx smaller) $\Rightarrow$ better approximation.
- For area under $f(x)$ on $[a, b]$: split into h bars of width $\Delta x = \frac{b-a}{h}$.
- Each bar has area (width) $\times$ (height) $= \Delta x \cdot f(\text{sample point})$.
- Left Riemann sum: $\Delta x \sum_{i=0}^{h-1} f(a + i \Delta x)$.
- Right Riemann sum: $\Delta x \sum_{i=1}^h f(a + i \Delta x)$.

Definite Integrals, Antiderivatives, Fundamental Theorem of Calculus

Definite Integrals

For a continuous function $f(x)$ on $[a, b]$, split into h equal bars with width $\Delta x = \frac{b-a}{h}$:

$$S_{\text{left}}(h) = \Delta x \sum_{i=0}^{h-1} f(a + i \Delta x), \quad S_{\text{right}}(h) = \Delta x \sum_{i=1}^h f(a + i \Delta x).$$

As we use *more* bars (so Δx gets *smaller*), both sums approach the same number R :

$$\lim_{h \rightarrow \infty} S_{\text{left}}(h) = \lim_{h \rightarrow \infty} S_{\text{right}}(h) = R.$$

Intuition: sum the areas of thin rectangles; thinner bars $\Rightarrow$ better total.

Definite Integrals

Definition 1 (Definite Integral).

With the setup above, the common limit R is the **definite integral**:

$$R = \int_a^b f(x) dx.$$

- $\int$ looks like an “S”: a special sum of many tiny pieces.
- a and b mark the lower and upper limits (where the adding happens).
- $f(x)$ is the *integrand* (the height being summed).
- dx reminds us we’re summing over tiny widths as $\Delta x \rightarrow 0$.
- It’s *definite* because the limits a and b are fixed numbers.

Definite Integrals II

An Example: Evaluate $\int_0^1 x^2 dx$.

Divide $[0, 1]$ into h bars. Then $\Delta x = \frac{1}{h}$ and the right endpoints are $x_i = \frac{i}{h}$, $i = 1, \dots, h$.

Right Riemann sum:

$$R(h) = \sum_{i=1}^h \Delta x f(x_i) = \sum_{i=1}^h \frac{1}{h} \left(\frac{i}{h}\right)^2 = \frac{1}{h^3} \sum_{i=1}^h i^2 = \frac{1}{h^3} \cdot \frac{h(h+1)(2h+1)}{6}.$$

As h grows (bars get thinner), this sum approaches the true area.

Definite Integrals III

$$\int_0^1 x^2 dx = \lim_{h \rightarrow \infty} R(h) = \lim_{h \rightarrow \infty} \frac{h(h+1)(2h+1)}{6h^3} = \lim_{h \rightarrow \infty} \frac{1}{6} \left(2 + \frac{3}{h} + \frac{1}{h^2} \right) = \frac{1}{3}.$$

Fact used: $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$

Antiderivatives

In simple terms, an "Antiderivative" is the opposite of a derivative.

Definition 2 (Antiderivative).

Let $f(x)$ be a function, with the derivative $f'(x)$.

Let $f'(x) = g(x)$. Then the *antiderivative* of $g(x)$ is $G(x)$, where $G'(x) = g(x)$.

Examples:

- An antiderivative of $f(x) = x^2$ is $F(x) = \frac{1}{3}x^3 + c$
- An antiderivative of $g(x) = x^n$ is $G(x) = \frac{1}{n+1}x^{n+1} + c$
- An antiderivative of $h(x) = e^{2x}$ is $H(x) = \frac{1}{2}e^{2x} + c$
- An antiderivative of $p(x) = \frac{1}{x}$ is $P(x) = \ln|x| + c$

Fundamental Theorem of Calculus

Definition 3 ((Second) Fundamental Theorem of Calculus).

Let f be a real-valued function on a closed interval $[a, b]$ and F a continuous function on $[a, b]$, which is an antiderivative of f on (a, b) : $F'(x) = f(x)$.

If f is Riemann integrable on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Fundamental Theorem of Calculus: Discussion

The theorem is extremely useful (*fundamental!*).

What it says, in simple terms: differentiation and integration are opposite operations. The integral of f from a to b is the antiderivative at b minus the antiderivative at a .

Example.

Coming back to our previous example, the integral of $f(x) = x^2$ on the interval $[0, 1]$ is:

$$\int_0^1 x^2 dx = \left[\frac{1}{3}x^3 \right]_0^1 = \frac{1}{3}(1^3) - \frac{1}{3}(0^3) = \frac{1}{3}.$$

Example.

Another example:

$$\int_1^4 x dx = \left[\frac{1}{2}x^2 \right]_1^4 = 7.5.$$

Properties of Definite Integrals

- *Constants*

$$\int_a^b kf(x)dx = k \int_a^b f(x)dx$$

- *Additive Property*

$$\int_a^b (f(x) + g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx$$

- *Linear Functions*

$$\int_a^b (k_1f(x) + k_2g(x))dx = k_1 \int_a^b f(x)dx + k_2 \int_a^b g(x)dx$$

Properties of Definite Integrals II

- *Intermediate Values*

for $a \leq b \leq c$:

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

- *Limit Reversibility*

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

Ex: $\int_0^1 x^2 dx = \int_0^{0.5} x^2 dx + \int_{0.5}^1 x^2 dx = \left(\frac{1}{3}x^3\right)_0^{0.5} + \left(\frac{1}{3}x^3\right)_{0.5}^1 =$
 $\frac{1}{3}(0.125 - 0) + \frac{1}{3}(1 - 0.125) = \frac{1}{3}$

Indefinite Integrals

Indefinite Integrals

Definite integrals have limits for the integration (what we showed with a and b in the definition.) Indefinite integrals do not. Then, the antiderivative solution will have a constant, $+c$ added to them to account for this.

Definition 4 (Indefinite Integral).

The *indefinite integral* of $f(x)$ is given by:

$$\int f(x) dx := F(x) + c$$

Ex: $\int (1 - 3x)^{\frac{1}{2}} dx = -\frac{2}{9}(1 - 3x)^{\frac{3}{2}} + c$

Indefinite Integrals – Properties

- *Linearity (as for definite integrals):*

$$\int [a f(x) + b g(x)] dx = a \int f(x) dx + b \int g(x) dx$$

- *Antiderivative family:* If $F'(x) = f(x)$, then $\int f(x) dx = F(x) + C$; any two antiderivatives differ by a constant.

- *Constant function:* $\int c dx = cx + C$

- *Verification:* Differentiate your result; for definite integrals constants cancel:

$$\left[F(x) + C \right]_a^b = F(b) - F(a)$$

Special Cases

Special Integrals

- $\int x^a dx = \frac{1}{a+1} x^{a+1} + C$, where $a \neq -1$
- $\int \frac{1}{x-a} dx = \ln |x-a| + C$
- $\int e^{ax} dx = \frac{1}{a} e^{ax} + C$, where $a \neq 0$
- $\int a^x dx = \frac{1}{\ln a} a^x + C$, where $a > 0$ and $a \neq 1$

More Integration Techniques

- For *Integration by Substitution*, see pages 218-219;
- For *Integration by Parts*, see pages 219-220-221;
- For *Multidimensional Integrals*, see Ch6.5 (pg 250) of J. Gill, 2006.

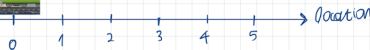
An Application: Inequality and the Gini Coefficient

- To be discussed in class.

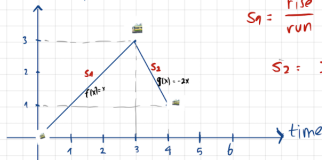
Appendix

Areas, Integrals : Train Example

Train



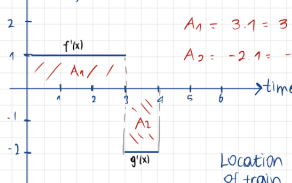
Location of train



$$S_1 = \frac{\text{rise}}{\text{run}} = \frac{3}{3} = 1 = f'(x)$$

$$S_2 = \frac{-2}{1} = -2 = g'(x)$$

Velocity of train



$$A_1 = 3 \cdot 1 = 3 = \int_0^3 f'(x) dx = \text{Change in Location of train from } t=0 \text{ to } t=3$$

$$A_2 = -2 \cdot 1 = -2 = \int_3^4 g'(x) dx = \text{Change in Location of train from } t=3 \text{ to } t=4$$

the (Riemann) integral of $g'(x)$ between 3 and 4

$$\text{Location of train at } t=4 = A_1 + A_2$$

$$= \int_0^3 f'(x) dx + \int_3^4 g'(x) dx$$